Supplemental material to this article can be found at:
http://dmd.aspetjournals.org/content/suppl/2011/06/01/dmd.111.039248.DC1.html

0090-9556/11/3909-1633–1642$25.00
DRUG METABOLISM AND DISPOSITION
Vol. 39, No. 9
39248/3707794
DMD 39:1633–1642, 2011
Printed in U.S.A.
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Physiologically Based Pharmacokinetic Modeling of Intestinal First-Pass Metabolism of CYP3A Substrates with High Intestinal Extraction[S]

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Received March 2, 2011; accepted June 1, 2011

ABSTRACT:

Prediction of intestinal availability (F_G), in conjunction with hepatic metabolism, is of considerable importance in drug disposition to assess oral clearance and liability to drug-drug interactions. In the current study, F_G predictions were performed within a physiologically based pharmacokinetic (PBPK) model using in vitro permeability and clearance data. The prediction success was assessed in comparison with the Q_{Gut} model. In addition, apparent oral clearance values, predicted using the PBPK model, were compared with in vivo observations from meta-analyses. Finally, unbound intrinsic clearance values (CLu_{int}) were determined for 12 CYP3A substrates in eight individual human jejunal microsome (HJM) samples to assess interindividual variability in intestinal intrinsic clearance and subsequent F_G predictions. Overall, the PBPK model improved F_G predictions in comparison with the Q_{Gut} model; this was apparent by a reduced bias and increased precision. In particular, F_G predictions of indinavir, saquinavir, and terfenadine were model-dependent. The predicted oral clearance values of the drugs investigated ranged from 8.79 to 6320 l/h for tacrolimus and simvastatin, respectively, and were overall within 3-fold of the observed data with the exception of indinavir, atorvastatin, and buspirone. The individual HJM CLu_{int} values ranged from 17 to 14,000 $\mu l \cdot min^{-1} \cdot mg^{-1}$ for atorvastatin and saquinavir, respectively, and corresponding interindividual variability in CLu_{int} estimates ranged from 41 to 67%. These in vitro data resulted in predicted F_G values ranging from 0.03 to 0.94 for simvastatin and indinavir, respectively. The largest interindividual variability of F_G was predicted for terfenadine (65%) in contrast with the low variability in the case of indinavir (3%).

Introduction

CYP3A enzymes represent the principle drug-metabolizing enzymes in the small intestine (Paine et al., 1997, 2006). The expression levels of CYP3A along the small intestine decline from the proximal to the distal regions (Paine et al., 1997; Zhang et al., 1999). A high drug concentration in the enterocytes during the absorption phase may lead to substantial metabolism despite the relatively low enzyme content (~1% in comparison to the liver) (Paine et al., 1997). On the other hand, it may lead to saturation or in some instances to inhibition of metabolism, resulting in nonlinear pharmacokinetics after oral drug administration. In addition, small intestinal metabolism is associated with substantial interindividual variability. For example, an up to 10-fold range in individual activity was demonstrated using metabolite formation data in the case of cyclosporine and tacrolimus (Lampen et al., 1995, 1996), which may partially reflect the high interindividual variability of CYP3A abundance (von Richter et al., 2004; Paine et al., 2006).

The importance of intestinal metabolism may be delineated from clinical studies after intravenous and oral drug administration or, alternatively, from grapefruit juice interaction studies under appropriate conditions. However, both in vivo methods are flawed by a number of inherent limitations discussed elsewhere (Galetin et al., 2008; Gertz et al., 2008a). The intestinal availability (F_G) for drugs such as cyclosporine, midazolam, felodipine, and tacrolimus is based on a considerable number of studies, in contrast to very limited and variable data for buspirone, statins, saquinavir, and terfenadine; hence, the in vivo F_G estimates for the latter drugs have to be regarded with caution. In addition, in vitro data may be used to make predictions of intestinal metabolism based on either in vitro clearance alone (Shen et al., 1997) or in combination with permeability data (Yang et al., 2007; Gertz et al., 2010). For the prediction of intestinal metabolism, the enterocytic, rather than the total intestinal blood flow, needs to be incorporated (Tam et al., 2003). The rationale for using enterocytic blood flow in the F_G predictions is that drugs pass through the enterocytes which contain the majority of metabolically active en-

This work was supported by a consortium of pharmaceutical companies (GlaxoSmithKline, Lilly, Novartis, Pfizer, and Servier) within the Centre for Applied Pharmacokinetic Research at the University of Manchester.

Article, publication date, and citation information can be found at http://dmd.aspetjournals.org.

doi:10.1124/dmd.111.039248.

[S] The online version of this article (available at http://dmd.aspetjournals.org) contains supplemental material.

ABBREVIATIONS: F_G, intestinal availability; Q_{Gut}, hybrid parameter of enterocytic blood flow and drug permeability; PBPK, physiologically based pharmacokinetic; HJM, human jejunal microsomes; LC, liquid chromatography; MS/MS, tandem mass spectrometry; MDCK-MDR1, Madin-Darby canine kidney cells transfected with human *MDR1* gene; CL, clearance; fu, fraction unbound; HIM, human intestinal microsomes; HLM, human liver microsomes; CV, coefficient of variation.

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zymes in the small intestine (Kolars et al., 1994). In general, no binding to enterocytic proteins is assumed, although this has not been satisfactorily investigated and is in contrast to the assumptions of the well stirred model for the liver. Advancement in computational power has allowed incorporation of intestinal transit times and heterogeneous expression levels of CYP3A into mechanistic predictions of intestinal absorption and metabolism. Complex models to predict intestinal absorption generally represent adaptations of the compartmental absorption and transit model (Yu and Amidon, 1999) and may account for drug dissolution, solubility, permeability, transport, and metabolism (Agoram et al., 2001; Tam et al., 2003; Jamei et al., 2009). Limiting the current trend toward holistic and mechanistic data interpretation is the need for extensive in vitro studies (e.g., full kinetic characterization of drug affinity for metabolizing enzymes and transporters) and formulation-specific data (e.g., dissolution and solubility) in conjunction with the quality of physiological information (e.g., abundance data of relevant enzymes and transporters) to support these models.

To reduce the necessary in vitro input data, previous work investigated F_G prediction success using the Q_{Gut} model, which simplifies prediction of intestinal metabolism to the input of in vitro clearance and permeability data (Yang et al., 2007; Gertz et al., 2010). The Q_{Gut} model has the basic structure of the well stirred model; however, the blood flow term is modified to represent a hybrid parameter, Q_{Gut}, consisting of physiological flow (i.e., enterocytic blood flow) and drug permeability (Chalasani et al., 2002; Rostami-Hodjegan and Tucker, 2002). However, our analysis highlighted considerable bias and imprecision in the prediction of F_G for drugs with in vivo F_G values less than 0.5 (Gertz et al., 2010). In particular, the Q_{Gut} model predictions of F_G for indinavir, saquinavir, and terfenadine were underpredicted by up to 96%. Thus, a physiologically based pharmacokinetic (PBPK) model was developed, which retains the use of in vitro clearance and permeability data but overcomes some of the limitations of the Q_{Gut} model. The PBPK model takes into account heterogeneity of metabolizing enzymes, drug concentration in the enterocytes, and any potential saturation of intestinal metabolism. The latter in particular was assumed to be responsible for some of the F_G underprediction trends observed previously. Prediction success of the PBPK model is assessed in comparison with the Q_{Gut} model using 12 CYP3A drugs with low intestinal availability in vivo. In addition, apparent oral clearance is predicted for the selected drugs using the PBPK modeling approach and compared with in vivo observations from meta-analyses. Finally, eight individual human jejunal microsome (HJM) samples are used to determine in vitro intrinsic clearance data of 12 selected drugs to assess interindividual variability in F_G predictions.

Materials and Methods

Determination of Intestinal Clearance In Vitro. In vitro clearance data were determined by substrate depletion in eight individual human jejunal microsomal samples prepared by an elution method from white donors. The microsomes from individual jejunal donors were purchased from BD Gentest (Woburn, MA) (Table 1). Substrate depletion experiments were performed in 0.1 M phosphate buffer (pH 7.4) containing 10 mM $MgCl_2$, 7.5 mM isocitric acid, 1.2 units/ml isocitric acid dehydrogenase, and 1 mM NADP. The microsomal protein concentrations ranged from 0.025 to 1.5 mg/ml for lovastatin/simvastatin and cyclosporine, respectively. The substrate concentrations were 10-fold below the reported K_m values in the literature for the drugs investigated. The drugs were added from methanol stock solutions, resulting in a final concentration of organic solvent in the incubation of 0.1% v/v. Clearance incubations were prepared as replicates of two in Eppendorf tubes at 37°C and 900 rpm in an Eppendorf Thermomixer. The metabolic reaction was initialized by the addition of NADP solution to the incubation mixture, and samples were taken at six designated time points within 60 min. Noncytochrome P450-dependent loss of drug over the incubation time was monitored by preparing additional samples in the absence of NADP. Metabolic reactions were terminated by removal of aliquots into an equal volume of ice-cold acetonitrile containing internal standard. Samples were centrifuged at 1000*g* for 20 min at 4°C in a Mistral 3001 centrifuge [MSE (UK) Ltd., London, UK], and 150 μl of supernatant was removed from each Eppendorf vial and transferred to glass vials before analysis on the LC-MS/MS system. Detailed information on LC-MS/MS analysis and the list of chemical suppliers have been reported in our previous publication (Gertz et al., 2010). Clearance values were corrected for experimentally determined nonspecific binding values to microsomal protein (fu_{inc}) (Gertz et al., 2008b, 2010). For cyclosporine, binding was predicted using drug lipophilicity data. The unbound intrinsic clearance values, CLu_{int}, were calculated using eq. 1. Clearance values were corrected for the mean population CYP3A abundance in the intestine of 50 pmol of CYP3A/mg protein (Paine et al., 2006) prior to F_G predictions.

TABLE 1

Summary of individual human jejunal microsome preparations

Donor[a]	Gender	Age	CYP3A Activity ± S.D.[b]
		yr	*nmol · min^{-1} · mg^{-1}*
HJM 1	Male	24	2.09 ± 0.176
HJM 2	Male	59	3.08 ± 0.085
HJM 3	Male	26	2.63 ± 0.151
HJM 4	Male	35	3.38 ± 0.485
HJM 5	Male	68	5.15 ± 0.313
HJM 6	Male	60	5.56 ± 0.122
HJM 7	Female	39	5.20 ± 0.198
HJM 8	Female	65	5.88 ± 0.393

[a] Microsome preparation by elution method from the jejunum.
[b] Testosterone 6β-hydroxylation activity was determined at a testosterone concentration of 250 μM.

$$CLu_{int} = \frac{V \cdot k}{protein_{microsomal}} \cdot \frac{1}{fu_{inc}} \quad (1)$$

where k represents the depletion rate constant, V represents initial incubation volume, and $protein_{microsomal}$ represents the initial amount of protein.

Testosterone 6β-Hydroxylation Activity. CYP3A enzyme activities of the HJM were determined at 250 μM testosterone concentration (substrate concentration sufficient to achieve V_{max} conditions) monitoring 6β-hydroxytestosterone formation over 10 min at a microsomal protein concentration of 0.5 mg/ml. Aliquots of 100 μl were removed into 100 μl of ice-cold acetonitrile containing progesterone (1 μM) as an internal standard. The appearance of 6β-hydroxytestosterone (m/z, 305.15 > 269.3, cone voltage 80 V, and collision energy 15 eV at maximum cone gas flow) was monitored. The LC-MS/MS system used consisted of a Waters 2795 with a Micromass Quattro Ultima triple quadruple mass spectrometer (Waters, Milford, MA). A Luna C18 column (3 μm, 50 × 4.6 mm) was used for chromatographic separation of the analytes using a gradient of acetonitrile and water containing 0.05% v/v formic acid. Composition of cofactors was the same as that in the substrate depletion assay; organic solvent content was 0.3% v/v methanol. Activity determination was performed on three separate occasions to assess interday variability.

Determination of K_m Values. Michaelis-Menten constants were determined for four drugs for which the literature indicated low and variable K_m values. Substrate depletion clearance was determined at the following substrate concentration ranges: felodipine, 0.5 to 10 μM (6 concentrations); indinavir, 0.025 to 5.5 μM (10 concentrations); nisoldipine, 0.25 to 5.0 μM (6 concentrations), and saquinavir, 0.05 to 5.0 μM (6 concentrations). The depletion profiles at the different substrate concentrations were performed in one representative microsomal preparation, HJM 6 (chosen for its high testosterone 6β-hydroxylation activity and low between-day variability). The CLu_{int} and K_m values together with associated S.E.s were determined using eq. 2 (Obach and Reed-Hagen, 2002). The equation reported by the original investigators was modified by multiplying both sides of the equation with the volume of incubation to transform depletion rate constants to intrinsic (at [S] approaching 0) and apparent intrinsic clearance (at any [S]). The fitting was performed in Grafit 5.0.10 (Erithacus Software Limited, Horley, Surrey, UK).

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$$CLu_{int,app} = CLu_{int}\left(1 - \frac{[S]}{[S] + K_m}\right) \quad (2)$$

where CLu_{int} represents the unbound intrinsic clearance, $CLu_{int,app}$ is apparent intrinsic clearance, $[S]$ is substrate concentration, and K_m is the Michaelis-Menten constant.

Preliminary Assessment of Enterocytic Drug Concentration and Possible Saturation of First-Pass Metabolism. Indinavir, saquinavir, and terfenadine F_G values were previously considerably underpredicted by 75 to 96% using the Q_{Gut} model (Gertz et al., 2010). Unlike the PBPK model, the Q_{Gut} model predictions did not take enterocytic concentrations (C_{ent}) into account. An initial assessment of enterocytic drug concentration was performed on the basis of eq. 3.

$$C_{ent} = \frac{D \cdot k_a \cdot F_a}{Q_{ent}} \quad (3)$$

The doses (D), absorption rate constants (k_a), and fractions absorbed (F_a) considered for the preliminary analyses are summarized in Table 2; an enterocytic blood flow (Q_{ent}) of 18 l/h was assumed. The selection of doses and formulations was based on the studies from which in vivo F_G estimates were obtained. The drugs investigated were ranked for their likelihood that saturation may have biased previous F_G estimates by assuming linearity in the Q_{Gut} model. From this analysis, a minor bias of intestinal metabolism can be anticipated by neglecting saturation of intestinal metabolism ($C_{ent}/K_m < 1$) for atorvastatin, felodipine, midazolam, and tacrolimus. In contrast, saturation was anticipated to affect the assessment of intestinal metabolism for cyclosporine, indinavir, saquinavir, and terfenadine to a greater extent ($C_{ent}/K_m > 10$). In addition, for indinavir, saturation of hepatic first-pass metabolism was anticipated. In fact, saturation of systemic clearance of indinavir (radiolabeled intravenous dose) was shown with concomitant administration of 400 to 800 mg of indinavir orally (Yeh et al., 1999). Equation 3 does not allow assessment of regional differences in C_{ent} and subsequent effects on F_G.

Reassessment of Atorvastatin F_G after Intravenous/Oral Administration. The in vivo estimate of atorvastatin F_G was reassessed on the basis of the available blood/plasma concentration ratio data ($R_b = 0.61$; Table 2). The initial assessment of atorvastatin F_G was based on the assumption of R_b being equivalent to 1 (Lennernäs, 2003), which resulted in a lower hepatic extraction ratio in comparison with the current analysis (0.42 versus 0.63) and subsequently a lower estimate of F_G (0.24 versus 0.38). The current F_G estimate of 0.38 assumes complete absorption of atorvastatin (Lennernäs, 2003). The reevaluation of atorvastatin F_G from intravenous/oral data compared favorably to data from grapefruit juice interaction studies (Gertz et al., 2008a).

Reassessment of Intrinsic Clearance of Cyclosporine and Tacrolimus. In comparison with Gertz et al. (2010), the unbound fractions in plasma of cyclosporine and tacrolimus were reevaluated from the available literature. Different methods have been used to determine both cyclosporine and tacrolimus fraction unbound in plasma. The current study favored fu_p measurements determined in stainless steel equilibrium dialysis chambers for cyclosporine (1.9%) and tacrolimus (1.3%), given their high reproducibility. The use of these fu_p values resulted in changed estimates of in vivo $CL_{int,h}$ in comparison to our previous publication (Table 2).

PBPK Model to Estimate F_G and CL_{oral} from In Vitro Data. A PBPK model was constructed in which tissues were connected by an arterial blood supply and a collective venous return to the lungs (Nestorov, 2003). All tissues were considered to be well stirred compartments, i.e., that the unbound tissue concentration is at equilibrium with the unbound concentration in the emergent blood (eq. 4).

$$V_T \cdot \frac{dC_T}{dt} = Q_T\left(C_{b,A} - \frac{C_T}{K_{b,T}}\right) \quad (4)$$

where V_T, C_T, Q_T, and $K_{b,T}$ represent the volume, concentration, blood flow, and tissue/blood concentration ratio of the different tissues and $C_{b,A}$ represents the arterial blood concentration.

The physiological values for blood flows and tissue volumes were taken from the literature (Brown et al., 1997; International Commission on Radiological Protection, 2002). The selected tissues accounted for >95% of total body weight; an additional compartment representing the rest of the body was included. Tissue/blood concentration ratios were either collated from the literature or predicted with mechanistic equations (Rodgers et al., 2005; Rodgers and Rowland, 2006) using human tissue composition data (Poulin and Theil, 2002). For tissues where no human data were available, rat tissue

TABLE 2

Summary of drug-related parameters used in the current PBPK model for 12 drugs investigated

References for $logP_{o:w}$, pK_a, fu_p, k_a, and K_m are found in Supplemental Table 1 and at http://www.pharmacy.manchester.ac.uk/capkr/.

	Dose	k_a[a]	C_{ent}	$logP_{o:w}$	pK_a	fu_p	R_b	P_{eff}[b]	K_m[c]	In Vivo $CL_{int,h}$[d]	In Vitro $CL_{int,h}$[e]
	mg	h^{-1}	*μM*					*μm/s*	*μM*	*l/h*	*l/h*
Atorvastatin	40	3.7	15	4.07	4.46	0.051	0.61	1.72	33	1990	213
Buspirone	20	5.0	14	2.63	7.32,4.12	0.05	0.81	7.17	8.0	20,700[f]	963
Cyclosporine[g]	380	2.0	32	3.45	Neutral	0.019[h]	1.36	3.30	1.4	1450	279
	570	1.1	14								
Felodipine	10	2.8	4.0	3.86	Neutral	0.0048	0.70	3.00	5.3	39,100	7150
Indinavir[g]	400	1.8	57	2.92	5.9, 3.7	0.36	0.84	2.07[i]	0.1	1120	2090
Lovastatin	20	0.8	9.0	4.26	Neutral	0.017	0.57	5.05	7.8	5750[f]	17,200
Midazolam	3	4.2	2.2	3.25	6.1	0.031	0.55	6.73	3.3	1610	1540
Nisoldipine	20	3.0	8.6	3.80	Neutral	0.0041	1.0	4.05	2.1	38,900	25,200
Saquinavir[g]	600	2.5	37	4.10	8.2	0.028	0.74	3.33[i]	0.3	7660[f]	26,800
Simvastatin	40	2.0	10	4.71	Neutral	0.014	0.57	4.30	3.4	14,300[f]	25,500
Tacrolimus	0.05 mg/kg	2.6	0.7	3.26	Neutral	0.013[h]	35	5.95[i]	2.6	7530	3750
Terfenadine	120	2.8	47	5.62	9.7	0.03	1.0	5.43[i]	1.0	70,600	12,400

$logP_{o:w}$, partitioning coefficient between octanol and water; $P_{app(A-B)}$, apparent permeability (apical-basolateral).

[a] Estimation of k_a was based on clinical data using

$$T_{max} = \frac{\ln\left(\frac{k_a}{k_{el}}\right)}{k_a - k_{el}}.$$

[b] Estimated from $P_{app(A-B)}$ data in MDCK-MDR1 cells (Gertz et al., 2010); for cyclosporine, in vivo P_{eff} data were used (Supplemental Table 1).
[c] Represent the unbound K_m values (if original studies reported microsomal protein concentration).
[d] Values reported in Gertz et al. (2010); updated fu_p values for atorvastatin, cyclosporine, felodipine, lovastatin, nisoldipine, simvastatin, and tacrolimus in comparison to Gertz et al. (2010).
[e] Data represent means from three human liver microsome pools (Gertz et al., 2010).
[f] In vivo intrinsic clearance was estimated from oral data; for buspirone intravenous clearance exceeded hepatic blood flow and showed very large interindividual variability (Gammans et al., 1986).
[g] Fraction absorbed of cyclosporine (Neoral and Sandimmune, 0.9 and 0.5, respectively), indinavir, and saquinavir (Invirase) 0.8 and 0.3, respectively.
[h] Based on equilibrium dialysis experiments in stainless steel chambers.
[i] Based on $P_{app(A-B)}$ data in the presence of a P-glycoprotein inhibitor.

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composition data were used. Concentration-time profiles, as well as F_G and CL parameters, were taken into consideration in the model development using midazolam and alfentanil as test compounds because of availability of K_b data in rat (data not shown). However, subsequent analysis focused on the prediction of F_G, $CL_{i.v.}$, and CL_{oral}.

The intestinal and liver compartments of this model are outlined in detail below, as they allowed the assessment of intestinal and hepatic availability and apparent intravenous and oral clearance. Systemic clearance was considered to occur exclusively in the liver (unless renal excretion was significant, e.g., for indinavir) and presystemic metabolism was considered to occur in both the small intestine and the liver. The liver was separated into liver tissue and the blood residing in the liver. The rate equations (eqs. 5 and 6) describe the change in concentrations in liver blood and liver tissue, respectively.

$$V_{b,Li} \cdot \frac{dC_{b,Li}}{dt} = C_{b,A} \cdot Q_{b,HA} + C_{b,PV} \cdot Q_{b,PV} - C_{b,Li} \cdot Q_{b,HV} - PS_{Li} \cdot (fu_b \cdot C_{b,Li} - fu_{Li} \cdot C_{Li}) \quad (5)$$

$$V_{Li} \cdot \frac{dC_{Li}}{dt} = PS_{Li} \cdot (fu_b \cdot C_{b,Li} - fu_{Li} \cdot C_{Li}) - \frac{V_{max} \cdot A_{CYP3A,Li} \cdot fu_{Li} \cdot C_{Li}}{K_m + fu_{Li} \cdot C_{Li}} \quad (6)$$

where $C_{b,A}$, $C_{b,PV}$, C_{Li}, and $C_{b,Li}$ represent the concentrations in the arterial blood, portal vein, liver, and hepatic outlet (or liver blood), respectively; $Q_{b,HA}$ (6.5% of cardiac output), $Q_{b,PV}$ (18.5% of cardiac output), and $Q_{b,HV}$ ($Q_{b,HA} + Q_{b,PV}$) represent the blood flows of the hepatic artery, portal vein, and hepatic vein, respectively; fu_b and fu_{Li} represent the fractions unbound in blood and liver ($fu_{Li} = fu_b/K_{b,Li}$ for drugs with no active uptake or efflux); $V_{b,Li}$ and V_{Li} represent the volumes of the blood residing in the liver and the liver tissue, respectively; PS_{Li}, V_{max}, and K_m represent the permeability-surface area product (10,000 times greater than hepatic blood flow to satisfy perfusion limited assumptions), the maximum metabolic velocity, and the Michaelis-Menten constant for metabolism, respectively; and $A_{CYP3A,Li}$ represents the total hepatic amount of CYP3A.

In the current analysis, no active uptake or efflux was considered to occur in the liver for the drugs under investigation. However, eqs. 5 and 6 can be modified to accommodate these processes by inclusion of the appropriate Michaelis-Menten or intrinsic clearance terms and accounting for the extracellular water fraction. The portal vein concentration represents the differential of emergent blood concentrations from the intestine (including the enterocytes), stomach, spleen, and pancreas.

The small intestine was divided into seven compartments (Fig. 1): 1, 2 to 3, and 4 to 7 represent the intestinal segments of duodenum, jejunum, and ileum, respectively (Yu and Amidon, 1999). The rate equations below describe the change of drug amount in the stomach and intestinal lumen (eqs. 7–10) and drug concentration in the enterocytes (eq. 11) with respect to time.

$$\frac{dA_{St}}{dt} = -A_{St} \cdot Kt_{St} \quad (7)$$

$$\frac{dA_{G,1}}{dt} = A_{St} \cdot Kt_{St} - Kt_{G,1} \cdot A_{G,1} - ka_{G,1} \cdot A_{G,1} \quad (8)$$

$$\frac{dA_{G,n}}{dt} = Kt_{G,n-1} \cdot A_{G,n-1} - Kt_{G,n} \cdot A_{G,n} - ka_{G,n} \cdot A_{G,n} \qquad n=2-7 \quad (9)$$

$$\frac{dA_{Co}}{dt} = A_{G,7} \cdot Kt_{G,7} - A_{Co} \cdot Kt_{Co} \quad (10)$$

$$V_{ent,n} \cdot \frac{dC_{ent,n}}{dt} = ka_{G,n} \cdot A_{G,n} - Q_{Gut,n} \cdot C_{ent,n} - \frac{V_{max} \cdot A_{CYP3Aent,n} \cdot fu_{ent,n} \cdot C_{ent,n}}{K_m + fu_{ent,n} \cdot C_{ent,n}} \quad (11)$$

where A denotes the amounts of drug in either the stomach (A_{St}), the intestinal segments ($A_{G,n}$), or the colon (A_{Co}); Kt_{St}, $Kt_{G,n}$ and Kt_{Co} refer to the transit rate constants of stomach, intestinal lumen, and colon, respectively; k_a represents the absorption rate constant determined using eq. 15; fu_{ent}, $C_{ent,n}$, $V_{ent,n}$, $A_{CYP3A,ent,n}$, and $Q_{Gut,n}$ refer to the unbound fraction, concentration, volume, amount of CYP3A, and the hybrid parameter of blood flow and permeability in the enterocytes of the nth intestinal compartment.

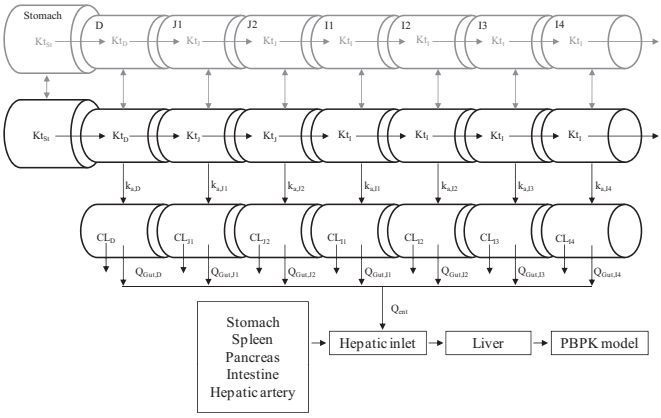


FIG. 1. Illustration of the absorption model applied for the current analyses. First (gray) and second levels represent undissolved and dissolved drug (after immediate release) in the different segments of the intestinal lumen, respectively; third level: enterocytes. St, stomach; D, duodenum; J1–2, jejunal segments; I1–4, ileal segments; *Kt*, transit rate constants; k_a, absorption rate constants; CL, apparent intrinsic clearance in the different intestinal segments, Q_{Gut}, hybrid function of regional enterocytic blood flows and drug permeability; Q_{ent}, total enterocytic blood flow.

Drug permeability was incorporated as a hybrid parameter of enterocytic blood flow and permeability, as in the Q_{Gut} model (Chalasani et al., 2002; Rostami-Hodjegan and Tucker, 2002). Absorption was considered to occur from the small intestinal compartments with the exception of saquinavir, for which colonic absorption was also incorporated, as investigated previously (Agoram et al., 2001). The blood flow to the small intestine represents approximately 10% of the cardiac output (39 l/h) (International Commission on Radiological Protection, 2002), and the enterocytic blood flow represents approximately 50% (i.e., 18 l/h) of the small intestinal blood flow (Granger et al., 1980). The cardiac output in the current assessment was 6.5 l/h, based on male subjects aged 20 to 35 years (Brown et al., 1997; International Commission on Radiological Protection, 2002). Effect of age on cardiac output (Brown et al., 1997) was taken into account for tacrolimus F_G predictions.

Intestinal availability, apparent intravenous, and oral clearance data were calculated from eqs. 12 and 13; where *t*-last represents a time appropriate to completely recover the administered dose as metabolites or drug excreted unchanged and $AM_{ent,n}$ represents the accumulative amount metabolized in the nth enterocyte compartment. The rate equations was solved in Matlab (version 7.10, 2010) using the ODE15s or 23s solvers. A mass balance equation was included in the script to allow monitoring of dose recovery over time.

$$CL_{app} = \frac{Dose}{\int_0^{t-last} C(t) \cdot dt} \quad (12)$$

$$F_G = 1 - \frac{\sum AM_{ent,n}}{Dose_{oral} \cdot F_a} \quad (13)$$

where $C(t)$ represents the drug concentration-time profile in blood or plasma, F_a represents fraction absorbed, and F_G represents the fraction of drug amount available to the enterocytes that enters the portal vein unchanged.

The main assumptions made in the current analysis are as follows: 1) there is no contribution of the small intestine to the systemic elimination of drugs; 2) there is no binding of drugs in the enterocytes (i.e., $fu_{ent} = 1$); 3) drug distribution into tissues satisfies the well stirred assumptions (no active uptake or efflux); 4) CYP3A is exclusively responsible for the metabolism of selected drugs; 5) dissolution and solubility do not affect assessment of F_G (special

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considerations for cyclosporine, indinavir, saquinavir, and terfenadine are outlined below); and 6) absorption occurs from the seven compartments of the small intestine only, with the exception of saquinavir for which colonic absorption was also incorporated.

Parameters for PBPK Modeling. The metabolism in the liver was scaled using the standard human microsomal recovery of 40 mg/g, average liver weight of 21.4 g/kg, and CYP3A content of 155 pmol/mg microsomal protein. The total CYP3A contents in the duodenum, jejunum, and ileum were 9.7, 38.4, and 22.4 nmol, respectively (Paine et al., 1997). The regional weights of the enterocytes and the transit rate constants in the duodenum, jejunum, and ileum were 18.2 g and 4.3 h^{-1}, 65.8 g and 1.7 h^{-1}, and 38.3 g and 2.1 h^{-1}, respectively (Yu et al., 1996; Paine et al., 1997). Differential blood supply to the duodenum, jejunum, and ileum was accounted for (Jamei et al., 2009; Darwich et al., 2011).

The oral absorption rate constants (k_a) of the drugs investigated here were estimated from apparent permeability data determined in MDCK-MDR1 cells. These data were first converted to effective permeability, P_{eff}, using the regression analysis previously performed on a set of 20 drugs (eq. 14) (Gertz et al., 2010). The P_{eff} data were then used to estimate absorption rate constants by eq. 15 (Yu and Amidon, 1999). The radii of the different intestinal compartments (r_{SI}) ranged from 0.85 and 1.58 cm for the ileum and duodenum, respectively (default values in GastroPlus version 7).

$$\log P_{eff} = 0.829 \cdot \log P_{app(A\text{-}B)} - 1.30 \quad (14)$$

$$k_a = \frac{2 \cdot P_{eff}}{r_{SI}} \quad (15)$$

For cyclosporine, indinavir, saquinavir, and terfenadine, saturation of intestinal metabolism was considered highly likely on the basis of preliminary analysis. Drug solubility data were therefore incorporated for these drugs, because luminal solubility may limit drug concentration in the enterocytes. In those cases, luminal rate equations (eqs. 7–10) were expanded to include the dissolved and undissolved drug amounts (Hintz and Johnson, 1989), as exemplified in eqs. 16 and 17. The current model assumed immediate drug release, dissolution from spherical particles, a constant particle radius over time, and equality of dissolution and precipitation rate constants. The occurrence of supersaturation was allowed.

$$\frac{dA_{un,n}}{dt} = A_{un,n-1} \cdot Kt_{n-1} - A_{un,n} \cdot Kt_n - \frac{3D}{\rho \cdot r \cdot h} \cdot A_{un,n} \cdot \left(C_{S,n} - \frac{A_{dis,n}}{V_n} \right) \quad (16)$$

$$\frac{dA_{dis,n}}{dt} = A_{dis,n-1} \cdot Kt_{n-1} + \frac{3D}{\rho \cdot r \cdot h} \cdot A_{un,n} \cdot \left(C_{S,n} - \frac{A_{dis,n}}{V_n} \right) - A_{dis,n} \cdot Kt_n - A_{dis,n} \cdot k_{a,n} \quad (17)$$

where A_{un} and A_{dis} represent, respectively, the undissolved and dissolved amount in the intestinal compartment n (including stomach and colon); Kt_n and ka_n represent the transit rate and absorption rate constants of the nth compartment; D represents the diffusion coefficient; ρ is the density; r is the particle radius; h is the diffusion layer thickness; $C_{S,n}$ is drug solubility in the nth is luminal compartment; and V_n is luminal volume of the different intestinal segments (default values in GastroPlus version 7).

For these four drugs, measurements in fasted simulated small intestinal fluids were kindly provided by Pfizer (Pharmacokinetics, Dynamics and Metabolism group, Sandwich, UK); additional solubility profiles across different pH values, if available, were obtained from the literature (Supplemental Table 1). The diffusion coefficients were estimated from molecular weights, the diffusion layer thickness was considered to be equal to particle radius (Hintz and Johnson, 1989), and a density of 1.2 g/ml was used (default value in GastroPlus version 7 and SimCYP version 10). A summary of drug solubility and particle size used can be found in Table 3.

Michaelis-Menten constants for CYP3A metabolism were collated from the literature except for felodipine, indinavir, nisoldipine, and saquinavir, for which K_m data were determined in the current study. A summary of K_m data, physicochemical properties, and drug permeability and clearance data used in the current analysis is provided in Table 2.

TABLE 3

Drug solubility data obtained in fasted simulated small intestinal fluids and particle size for cyclosporine, indinavir, saquinavir, and terfenadine

Additional references are found in Supplemental Table 1.

	Solubility	Particle Radius
	μM	μm
Cyclosporine	9.1 (13.3)[a]	0.018,1.87[b]
Indinavir	90[c]	25[d]
Saquinavir	64[c]	25[d]
Terfenadine	38[e]	25[d]

[a] Value in parentheses represents fasted human intestinal fluids.
[b] Cyclosporine (Neoral and Sandimmune, respectively).
[c] Refer to indinavir sulfate and saquinavir mesylate; solubility data of indinavir and saquinavir over pH range were collated from the literature and the maximal solubility was limited to the highest reported value.
[d] Assumed.
[e] Henderson-Hasselbalch equation for a monoprotic base was used to estimate pH dependent solubility; maximum solubility was limited to 848 μM.

In vitro intrinsic clearance data used for predictions were determined in either pooled intestinal and liver microsomes (Gertz et al., 2010) or eight individual HJM reported in the current study. The clearance data obtained in the intestinal and liver pools were used for a comparison of model performance between the F_G predictions by the Q_{Gut} and the PBPK model and for the prediction of apparent intravenous and oral clearance using the PBPK model. On the other hand, the individual HJM data were used to assess interindividual variability in F_G predictions using the PBPK model alone.

Measurement of Prediction Success and Comparison with In Vivo Data and Q_{Gut} Predictions. The predictions of F_G and apparent intravenous and oral clearance data were compared with corresponding in vivo data summarized in Table 4. Bias and precision of F_G and clearance predictions were assessed by geometric fold error and root mean squared error.

Results

Small intestinal metabolism was assessed for 12 drugs in eight individual human jejunal microsomal preparations (six male and two female white donors) (Table 1). The mean CYP3A activity was 4.12 nmol · min^{-1} · mg^{-1}, ranging from 2.09 to 5.88 nmol · min^{-1} · mg^{-1} as measured by formation of 6β-hydroxytestosterone at 250 μM testosterone concentration. The mean CYP3A activity was associated with a coefficient of variation of 36%; a 2.8-fold difference was observed between the donors with the lowest and highest CYP3A activity. The between-day variability in testosterone 6β-hydroxylation activity determinations was low, ranging from 2 to 14%.

The individual intrinsic clearance values determined in the current study showed a more than 800-fold difference, ranging from 17.0 to 14,000 μl · min^{-1} · mg^{-1} for atorvastatin and saquinavir, respectively (Fig. 2). HJM 1 showed the lowest CYP3A activity and resulted, on average, in the lowest intrinsic clearance values for the drugs investigated. In contrast, HJM 6 generally resulted in the highest clearance values while displaying the second highest CYP3A activity. The mean HJM intrinsic clearance values ranged from 33.5 to 7220 μl · min^{-1} · mg^{-1} for atorvastatin and saquinavir, respectively, whereas the coefficient of variation ranged from 41 to 67% for simvastatin and tacrolimus, respectively (Table 4). Overall, the testosterone 6β-hydroxylation activity was a reasonable predictor for intrinsic clearance values between different individual jejunal microsomal preparations ($R^2 = 0.72$ combined data of all 12 substrates). The intrinsic clearance values of atorvastatin and saquinavir were poorly correlated with testosterone 6β-hydroxylation activity ($R^2 < 0.30$), in contrast to the good correlation seen for buspirone, midazolam and simvastatin ($R^2 > 0.85$).

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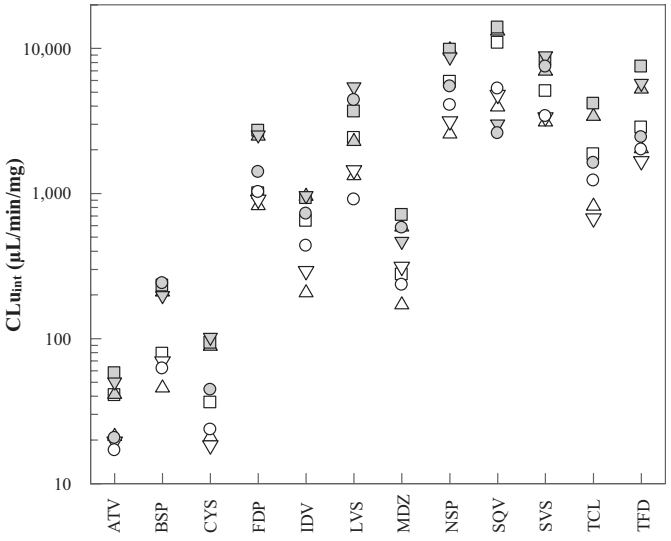


FIG. 2. Individual unbound intrinsic clearance values from eight HJM preparations for atorvastatin (ATV), buspirone (BSP), cyclosporine (CYS), felodipine (FDP), indinavir (IDV), lovastatin (LVS), midazolam (MDZ), nisoldipine (NSP), saquinavir (SQV), simvastatin (SVS), tacrolimus (TCL), and terfenadine (TFD); HJM 1 (△), HJM 2 (□), HJM 3 (▽), HJM 4 (○), HJM 5 (▲), HJM 6 (▨), HJM 7 (▼), and HJM 8 (●); for tacrolimus, CLu_{int} data in HJM preparation 7 were not included, as the no-NADP incubation showed extensive drug depletion.

To assess the K_m values of indinavir, felodipine, nisoldipine, and saquinavir, intrinsic clearance values were measured via a substrate depletion approach across a range of substrate concentrations. The K_m values were estimated using eq. 2 at 0.1 ± 0.01 (mean and SE from the fit), 0.3 ± 0.02, 2.1 ± 0.3, and 5.3 ± 0.4 μM for indinavir, saquinavir, nisoldipine, and felodipine, respectively.

Mechanistic Predictions of F_G and Comparison to the Q_{Gut} Model. The predictions of intestinal first-pass metabolism were based on the PBPK model outlined under *Materials and Methods*. Differential CYP3A amounts along the small intestine, intestinal transit time, drug-specific absorption rates, and differential blood flows to the intestinal segments were taken into account. Furthermore, the inclusion of Michaelis-Menten kinetic parameters allowed the assessment of potential nonlinear first-pass metabolism in small intestine and liver.

TABLE 4

Individual CLu_{int} values of 12 drugs determined by a substrate depletion method in eight individual HJM preparations and comparison to previously published data in human intestinal and hepatic microsomal pools

	CLu_{int}			CV[a]
	HIM[b]	HLM[b]	HJM Mean[a]	
	$\mu l \cdot min^{-1} \cdot mg^{-1}$			%
Atorvastatin	13.6	59.3	33.5	47
Buspirone	108	268	142	60
Cyclosporine	27.7	79.5	53.5	66
Felodipine	1170	1990	1610	51
Indinavir	298	582	645	47
Lovastatin	2440	4790	2740	59
Midazolam	340	429	418	47
Nisoldipine	3840	7000	6180	47
Saquinavir	3030	7460	7220	65
Simvastatin	3480	7100	5840	41
Tacrolimus	658	1040	1970	67
Terfenadine	1650	3440	3690	59

[a] Data from the present study; for tacrolimus, CLu_{int} value represents the mean and CV of seven individual intrinsic clearance values.

[b] CLu_{int} data from HIM pool (n = 10 donors) and three HLM pools (n = 105 donors in total) reported in Gertz et al. (2010).

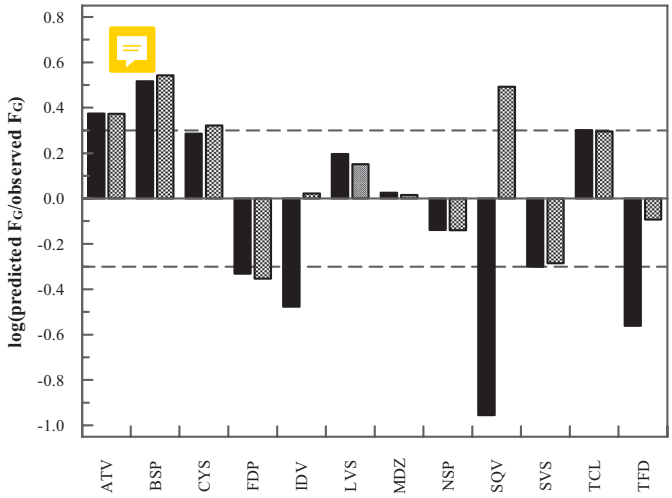


FIG. 3. Drug-specific F_G prediction success using the Q_{Gut} (Gertz et al., 2010) or the PBPK model (■ and ▨, respectively) of atorvastatin (ATV), buspirone (BSP), cyclosporine (CYS), felodipine (FDP), indinavir (IDV), lovastatin (LVS), midazolam (MDZ), nisoldipine (NSP), saquinavir (SQV), simvastatin (SVS), tacrolimus (TCL), and terfenadine (TFD); dashed lines indicate a 2-fold deviation from unity. In vitro clearance and permeability data used are listed in Table 2.

The F_G predictions obtained by the PBPK model were compared with the predictions obtained by the Q_{Gut} model (Fig. 3). To allow a direct comparison between the models, our previously published in vitro intrinsic clearance and permeability data were used (Table 5). The F_G predictions of atorvastatin, felodipine, midazolam, nisoldipine, and simvastatin were marginally affected by the choice of model. However, considerable differences were observed in the F_G predictions of indinavir, saquinavir, and terfenadine for which the PBPK model predicted higher F_G values, more in line with the observed data. The F_G predictions for indinavir, saquinavir, and terfenadine using the PBPK model were 0.98, 0.56, and 0.32, respectively. Differences in the F_G estimates were due to the ability of the PBPK model to account for saturation of intestinal metabolism by using V_{max} and K_m data rather than CL_{int}. In contrast to indinavir, the predicted F_G values of saquinavir and terfenadine were very sensitive to changes in the K_m values used. A reduction in prediction accuracy was observed for buspirone and cyclosporine when the PBPK model was compared with the Q_{Gut} model. For these drugs, F_G overprediction observed using the Q_{Gut} model increased further in the PBPK model (e.g., for cyclosporine from 93 to 110% overprediction). For tacrolimus, a reduction in enterocytic blood flow in response to a reduction in cardiac output with age resulted in lower F_G estimates than those reported previously (clinical data of tacrolimus were generally reported in individuals aged 50 or older). This result was accurately accounted for by both models. An overall improvement in F_G predictions was observed using the PBPK model as measured by the decrease in geometric mean fold error (from 2.4 to 1.8) and root mean squared error value (from 0.32 to 0.28) in comparison with those for the Q_{Gut} model.

Prediction of Apparent Oral Clearance. In contrast with the Q_{Gut} model, the PBPK model also allowed the prediction of apparent oral clearance data of the 12 drugs investigated. Analogous to the F_G predictions, this analysis was performed using the mean in vitro intrinsic clearance data from the HIM and HLM pools summarized in Table 4. The predicted oral clearance values are summarized in Table 5 and Fig. 4. The in vivo oral clearance data ranged from 20.7 to 14,400 l/h for tacrolimus and saquinavir, respectively; for cyclosporine and tacrolimus, observed oral clearance values were obtained from blood data. Prediction success of apparent oral clearance inside 3-fold

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TABLE 5

Comparison of predicted F_G and apparent intravenous and oral clearance data for 12 drugs and observed data

Predictions were performed using either the Q_{Gut} or PBPK modeling approach.

	Observed			Predicted			
	F_G[a]	$CL_{i.v.}$[a]	CL_{oral}[a]	F_G (Q_{Gut})[b]	F_G (PBPK)[b]	$CL_{i.v.}$[b]	CL_{oral}[b]
		l/h					l/h
Atorvastatin	0.38[c]	37.5	949	0.90	0.90	9.05[d]	12.0[d]
Buspirone	0.21	119	4910	0.69	0.73	29.7	70.6
Cyclosporine							
Neoral	0.44	16.8[e]	51.7[e]	0.85	0.92	3.71[e]	17.1[e] (F_a = 24%)
Sandimmune	0.44	16.8[e]	107[e]	0.85	0.92	3.70[e]	24.6[e] (F_a = 16%)
Felodipine	0.45	50.0	462	0.21	0.20	22.5	254
Indinavir	0.93	77.3	59.6	0.31	0.98	82.0	547 (F_a = 61%)
Lovastatin	0.07		1380	0.11	0.10	45.9	4110
Midazolam	0.51	25.9	102	0.54	0.53	24.7	150
Nisoldipine	0.11	60.5	1340	0.08	0.08	49.0	1990
Saquinavir	0.18	54.2	14,400	0.02	0.56[f]	65.6	3620[f] (F_a = 41%)
Simvastatin	0.14		1630	0.07	0.07	48.0	6320
Tacrolimus	0.14	2.69[e]	20.7[e]	0.28	0.28	1.36[e]	8.79[e]
Terfenadine	0.40		5290	0.11	0.32	76.9	2720 (F_a = 52%)

[a] Summary from Gertz et al. (2010).
[b] Predictions were performed using the clearance data from HIM and HLM and apparent permeability data determined in MDCK-MDR1 cells reported previously (Gertz et al., 2010).
[c] Reassessed from the intravenous clearance data provided in Lennernäs (2003) and R_b value of 0.61 (see *Materials and Methods*).
[d] Acid-lactone conversion was not considered. Predictions based on 70-fold higher intrinsic clearance values of the lactone (Jacobsen et al., 2000) resulted in apparent intravenous and oral clearance values of 54.4 and 786 l/h, respectively.
[e] Apparent blood clearance values.
[f] Absorption in the proximal colon (Agoram et al., 2001) was incorporated in the current model.

of unity was observed for cyclosporine (Neoral), felodipine, lovastatin, midazolam, nisoldipine, tacrolimus, and terfenadine. For cyclosporine, indinavir, saquinavir, and terfenadine, incomplete absorption was predicted using the drug solubility data (Table 5).

Due to a combination of underprediction of hepatic and intestinal clearance, a particularly low prediction success of oral clearance was observed for atorvastatin and buspirone (both <2% of the observed value). In contrast, overpredictions of oral clearance were observed for indinavir and simvastatin. These were due to an approximately 2-fold overprediction of hepatic intrinsic clearance (Table 2) for both drugs. In addition, for indinavir, the incorporation of drug solubility data predicted incomplete absorption (61%), which contributed to the overprediction of CL_{oral}. Saturation of hepatic first-pass metabolism of indinavir, on the other hand, was predicted by the model, as illustrated by the ratio of the areas under the curve in hepatic out- and inlet (F_H = 32 versus 11% under linear conditions). For felodipine, overprediction of intestinal metabolism (observed and predicted F_G values were 0.45 and 0.20, respectively) and underprediction of apparent hepatic clearance (Table 5) resulted in underprediction of apparent CL_{oral} by 1.8-fold. Overall, the average bias was 4.7-fold for the predictions of apparent oral clearance, largely driven by the extensive underpredictions for buspirone and atorvastatin CL_{oral}; removal of these two drugs from the dataset reduced the bias to 2.9-fold.

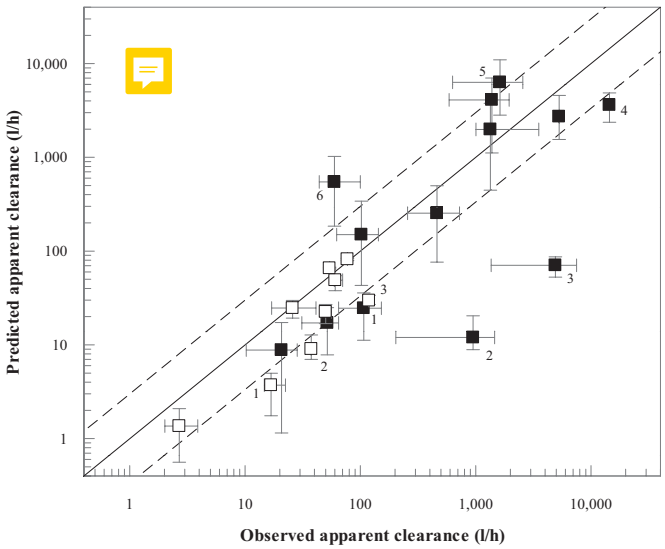


FIG. 4. Comparison of observed and predicted apparent clearance after intravenous (□) and oral (■) administration. In all instances data represent plasma clearance, with the exception of cyclosporine and tacrolimus (blood clearance). Predictions were performed using the in vitro data (Table 2) and the current PBPK model. The clinical data were determined by meta-analyses (Gertz et al., 2010); the error bars on the *x*- and *y*-axis indicate the range in the clinical data and in the predictions using different microsomal pools, respectively. The dashed lines indicate a 3-fold deviation from unity; outliers are highlighted: 1, cyclosporine; 2, atorvastatin; 3, buspirone; 4, saquinavir; 5, simvastatin; and 6, indinavir.

Predictions of F_G Values from Individual Human Jejunal Microsomes. F_G predictions were also performed using individual data from eight human jejunal microsomal preparations (Fig. 5). In accordance with the generally higher in vitro clearance data obtained in the individual HJM in comparison to the HIM pool (Table 4), the predicted F_G values were overall lower than those obtained from data in pooled microsomes. Mean F_G predictions ranged from 0.03 to 0.94 for simvastatin and indinavir, respectively (Table 6). The coefficient of variation in F_G predictions from the individual HJM clearance data ranged from 3 to 65% for indinavir and terfenadine, respectively. The F_G predictions of terfenadine showed the highest CV of all drugs investigated, resulting in an 8.5-fold difference between the lowest and highest estimate. The remaining drugs showed less than 5-fold differences with the exception of lovastatin and tacrolimus (both 5.5-fold). The lowest F_G values (~2%) were predicted for nisoldipine from HJM preparations 5, 6, and 7, which also showed the highest intrinsic clearance values of nisoldipine (Figs. 2 and 5). Higher CLu_{int} values were determined for saquinavir; however, these did not result in lower F_G values because of the prediction of saturation of intestinal first-pass for this drug. The highest F_G values and lowest CV were predicted for indinavir (0.92–0.98). This result was not due to low metabolic clearance, the average CLu_{int} value of indinavir being 645 $\mu l \cdot min^{-1} \cdot mg^{-1}$ (greater than midazolam intrinsic clearance) but

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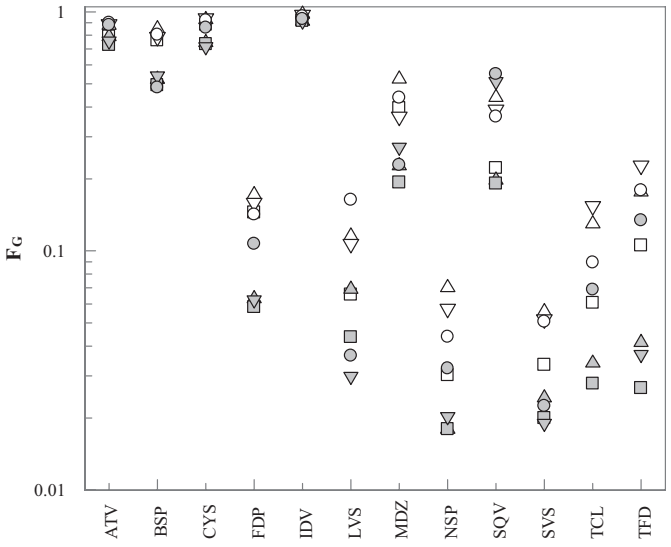


FIG. 5. Individual F_G predictions using the mechanistic first-pass model and intrinsic clearance data determined in eight individual HJM preparations (Fig. 2) for atorvastatin (ATV), buspirone (BSP), cyclosporine (CYS), felodipine (FDP), indinavir (IDV), lovastatin (LVS), midazolam (MDZ), nisoldipine (NSP), saquinavir (SQV), simvastatin (SVS), tacrolimus (TCL), and terfenadine (TFD): HJM 1 (△), HJM 2 (□), HJM3 (▽), HJM 4 (○), HJM 5 (▲), HJM 6 (▨), HJM 7 (▼), and HJM 8 (●).

was due to the prediction of extensive saturation of intestinal metabolism. Consistent with the predictions based on the data from microsomal pools, considerable F_G overpredictions were apparent for atorvastatin and buspirone from the individual microsomal preparations (2–3-fold, respectively).

Intrinsic clearance data from eight individual HJM were used to assess interindividual variation of F_G. In the literature, very few clinical studies report individual data to allow an estimation of interindividual variability of this parameter. For felodipine, individual area under the curve data before and after grapefruit juice coadministration have been reported in four separate studies, for a total of 45 individuals, summarized in Gertz et al. (2008a). The mean F_G of felodipine was 0.45; this estimate was associated with a between-individual variability of 41% and ranged from 0.12 to 1.0 (8.3-fold between highest and lowest F_G). By using the HJM clearance data, the predicted F_G of felodipine was 0.11 (range 0.06–0.17), and the CV of 42% reflected the variability observed in vivo. In the case of midazolam, individual data after intravenous and oral dose have been reported in 20 individuals (Thummel et al., 1996). This study indicated a 41% interindividual variability of F_G, which ranged from 0.21 to 0.94 (4.6-fold ratio between highest and lowest F_G). By using the HJM clearance data, the predicted F_G of midazolam was 0.33 (range 0.19–0.52) with a CV of 36%, which reflected the variability observed in vivo.

Discussion

Determination of Intestinal Clearance In Vitro. The in vitro clearance values from eight individual human jejunal microsomal samples compared well with our previous data generated in pooled human intestinal microsomes (Table 4). The CYP3A activity was considerably greater than that in the HIM pool, despite the fact that elution methods were used in both cases to obtain the microsomes (1.84 and 4.12 nmol · min^{-1} · mg^{-1} for HIM pool and HJM, respectively). The mean activity of the HJM was comparable to the mean of the three HLM pools (4.65 nmol · min^{-1} · mg^{-1}). The between-individual variability of 36% was lower than anticipated from reports on tacrolimus and cyclosporine in 14 small intestinal samples, for which the between-subject variability was 54 to 70%, a 4.6- to 11-fold difference between highest and lowest activity in contrast to 2.8-fold in the current study (Lampen et al., 1995, 1996). Considering the relatively small sample size, one can question whether the current HJM selection accurately reflects the true population variability of intestinal CYP3A activity. However, to our knowledge this is the largest dataset of intrinsic clearance data obtained in individual intestinal microsomes in the current literature.

Prediction of F_G Using PBPK Modeling. In this study, a PBPK model was applied to make predictions of intestinal availability using in vitro clearance and permeability data for 12 selected drugs. The analysis focused on drugs with high intestinal extraction in vivo, because F_G prediction accuracy for these drugs based on the Q_{Gut} model was lower that for drugs with $F_G > 0.5$ (Gertz et al., 2010). Use of the PBPK model improved F_G prediction, resulting in lower bias and increased precision. In contrast to the Q_{Gut} model, the PBPK model accounted for substrate concentrations at the enzyme site, Michaelis-Menten constants, and potential saturation of metabolism. In addition, the PBPK model accommodated any potential regional differences in intestinal availability because of the heterogeneous expression of CYP3A enzymes along the small intestine (Paine et al., 1997). Solubility of cyclosporine, indinavir, saquinavir, and terfenadine in simulated small intestinal fluids under fasted conditions was taken into account; for cyclosporine, these data compared well with reports in actual human intestinal fluids (Table 3). Because of a lack of transporter-specific kinetic data (K_m and V_{max} and regional abundance data for key transporters in the small intestine), the current model did not account for active uptake or efflux in either the small intestine or the liver, which may contribute to the absorption and disposition of a number of drugs in the current dataset. At present, even for P-glycoprotein, the most studied transporter in the small intestine, conflicting reports regarding its distribution exist in the literature. Data are generally based on either mRNA levels (Nakamura et al., 2002; Thörn et al., 2005; Berggren et al., 2007), or, if protein levels are determined, on a small sample size ($n = 4$) (Mouly and Paine, 2003) or on comparison of regional preparations from different individuals (Berggren et al., 2007). We refrained from performing a sensitivity analysis on the interplay of CYP3A and P-glycoprotein, as a publication dedicated exclusively to that effect has been presented (Darwich et al., 2011).

In addition to the need for more informative data on intestinal transporters, refinement of regional cytochrome P450 contents based

TABLE 6

Summary of the average, range, and CV of F_G values predicted from individual clearance data determined in eight human jejunal microsomal preparations using the PBPK modeling approach

	F_G	Range	CV
			%
Atorvastatin	0.83	0.73–0.90	8
Buspirone	0.65	0.48–0.85	24
Cyclosporine	0.84	0.71–0.94	11
Felodipine	0.11	0.06–0.17	42
Indinavir[a]	0.94	0.92–0.98	3
Lovastatin	0.08	0.03–0.16	59
Midazolam	0.33	0.19–0.52	36
Nisoldipine	0.04	0.02–0.07	53
Saquinavir[a]	0.36	0.19–0.55	39
Simvastatin	0.03	0.02–0.06	45
Tacrolimus[a]	0.08	0.03–0.15	58
Terfenadine[a]	0.12	0.03–0.23	65

[a] Based on apparent permeability (apical-basolateral) data in the presence of a P-glycoprotein inhibitor.

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on a larger number of individuals and preferably determined by a less destructive preparation technique will provide greater confidence in the predictions of intestinal metabolism. The current data presented by Paine et al. (1997) are based on a mucosal scraping technique and a relatively small sample size, in contrast to hepatic CYP3A content data obtained from a meta-analysis of a large number of individuals (Rostami-Hodjegan and Tucker, 2007).

Prediction of Apparent Intravenous and Oral Clearance. Successful predictions of apparent oral clearance (inside 3-fold) were obtained using pooled microsomal intrinsic clearance data for 6 of 12 drugs. In contrast, large underpredictions of oral clearance were observed for atorvastatin and buspirone (consistent with underprediction of $CL_{i.v.}$). The metabolism of buspirone has been suggested to be mainly mediated by CYP3A and a similar metabolite pattern in HLM was reported in comparison with that in vivo (Jajoo et al., 1989; Zhu et al., 2005). Previously reported metabolite formation data resulted in marginally higher CLu_{int} values (correction for fu_{inc} applied) compared with our substrate depletion data (460 versus 268 $\mu l \cdot min^{-1} \cdot mg^{-1}$) (Zhu et al., 2005; Gertz et al., 2010). Buspirone clinical data after intravenous administration in eight volunteers were highly variable (i.e., $CL_p = 28.3 \pm 10.3$ ml $\cdot$ min^{-1} $\cdot$ mg^{-1}, $F = 3.9 \pm 4.3\%$) with negligible renal excretion of unchanged drug (Gammans et al., 1986). However, neither the variability in the in vivo data nor the differences in the in vitro data are sufficient to explain the extensive underprediction of intrinsic hepatic clearance, because the in vitro data would classify buspirone as a low to moderate extraction drug ($E_H = 38\%$; Table 5), whereas the in vivo data show clearance values in excess of hepatic blood flow.

A number of factors may contribute to the extensive underprediction of both intravenous and oral clearance of atorvastatin. First, in vitro data have highlighted drug affinity for uptake transporters (Kameyama et al., 2005) and studies in polymorphic organic anion-transporting polypeptide 1B1 populations confirmed the clinical relevance of at least organic anion-transporting polypeptide 1B1 to atorvastatin disposition (Pasanen et al., 2007). It has been suggested that interconversion between acid and lactone forms may represent the initial step in atorvastatin metabolism (Jacobsen et al., 2000); this may occur directly from the parent or from the acyl glucuronide metabolite (Prueksaritanont et al., 2002). In human liver microsomes, the lactone displays on average a 70-fold greater intrinsic clearance than the hydroxy acid form (Jacobsen et al., 2000). Incorporation of the higher intrinsic clearance of the lactone form into the current analysis resulted in apparent intravenous and oral clearance values of 54.4 and 786 l/h, respectively, and better agreement with the observed data (37.5 and 949 l/h, respectively). The acid-lactone conversion was assumed to occur in plasma and therefore only CLu_{int} for hepatic metabolism was modified in the PBPK model, whereas intestinal intrinsic clearance remained unaltered. However, given the large variability in the ratio of CL_{int} of the lactone and the acid form (14- to 160-fold reported from four individual HLM) (Jacobsen et al., 2000), more data are required to investigate the rate-limiting step in atorvastatin metabolism.

In the case of cyclosporine, the reason for an ~5-fold underprediction of intrinsic clearance is unclear. Although UDP-glucuronosyltransferase-mediated metabolism in HLM has been reported (Strassburg et al., 2001), we were unable to confirm this by a substrate depletion assay performed in alamethicin-activated microsomes in the absence and presence of bovine serum albumin (data not shown). The underestimation of $CL_{int,\,h}$ of cyclosporine was masked in the reasonably well predicted oral clearance by the fact that cyclosporine absorption was also underestimated (F_a of 24 and 16% for Neoral and Sandimmune, respectively).

F_G Predictions Using Individual Jejunal Microsomal Samples. Intrinsic clearance data from eight individual HJM were used to predict F_G and associated interindividual variability. Individual in vivo F_G values were reported for terfenadine, tacrolimus, midazolam, and felodipine in 6, 12, 20, and 45 individuals, respectively. Use of individual HJM clearance data in the PBPK model resulted in an underprediction of felodipine F_G (consistent with the pooled data); however, the predicted variability in F_G reflected the variability seen in vivo. Considering extensive binding of felodipine to plasma proteins (>99%) (Table 2), binding to enterocytic proteins during absorption cannot be ruled out. Any binding to the enterocytic proteins would lead to reduced apparent intestinal clearance and therefore increased F_G predictions. Midazolam F_G was well predicted using the pooled microsomal CLu_{int} data but underpredicted using the individual HJM data. Although the coefficient of variation was predicted reasonably well, the F_G range was underpredicted. In particular, the ability to predict the upper limit of in vivo F_G values was poor, suggesting that selected individual jejunal donors did not cover the full spectrum of CYP3A activity present in the small intestine in vivo.

Other examples for which individual estimates of F_G were available in the literature include tacrolimus (Floren et al., 1997; Hebert et al., 1999) and terfenadine (Clifford et al., 1997). The interindividual variabilities of in vivo F_G estimates for tacrolimus and terfenadine were 36 and 54%, respectively. Both the CV and the fold difference between the highest and lowest F_G values were overpredicted for terfenadine. In addition, the mean predicted F_G of terfenadine was underestimated using the individual HJM data. Extensive binding to microsomal proteins has been observed for terfenadine using equilibrium dialysis (Gertz et al., 2008b). Any inaccuracies in the fu_{inc} estimate will bias the estimate of intrinsic clearance and also the estimate of unbound K_m used in the PBPK model. In contrast, prediction of the mean tacrolimus F_G was more successful (0.08 versus 0.14 for the predicted and observed F_G, respectively); however, both the CV and the fold difference were overpredicted using the intrinsic clearance data from eight individual HJM (CV = 58% and 5.5-fold). As an alternative, a propagation of the variability in CYP3A in the small intestine (von Richter et al., 2004) using Monte Carlo simulations may be used to predict interindividual variability of intestinal metabolism and F_G. However, it has to be emphasized that CYP3A abundance of a larger population needs to be assessed to capture the interindividual variability correctly.

In conclusion, the use of a PBPK model to study intestinal metabolism represents an improvement over the previously reported F_G predictions of high-extraction drugs using the Q_{Gut} model. Accounting for drug concentration and the region of absorption plays an important role in the assessment of intestinal metabolism and can be propagated into assessment of potential drug-drug interactions. In addition to F_G predictions, the PBPK model allowed the assessment of apparent intravenous and oral clearance, with the majority of the drugs predicted within 3-fold of the observed data. Finally, the contribution of intestinal transporters or metabolizing enzymes other than CYP3A needs to be integrated in the models; currently, this is limited by the general lack of unambiguous abundance data in tandem with drug-specific in vitro kinetic data.

Acknowledgments

We thank Sue Murby for technical assistance with LC-MS/MS analyses and Professor Leon Aarons and Drs. Kayode Ogungbenro and Henry Pertinez for useful discussions on the PBPK model (all University of Manchester, Manchester, UK). Additional thanks go to Suet Mei Wong at Pfizer, Pharmacokinetics, Dynamics and Metabolism group (Sandwich, UK) for the determination of solubility data in fasted simulated small intestinal fluids.

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Authorship Contributions

Participated in research design: Galetin, Gertz, and Houston.

Conducted experiments: Gertz.

Performed data analysis: Gertz.

Wrote or contributed to the writing of the manuscript: Gertz, Galetin, and Houston.

Other: Galetin and Houston.

References

Agoram B, Woltosz WS, and Bolger MB (2001) Predicting the impact of physiological and biochemical processes on oral drug bioavailability. *Adv Drug Deliv Rev* **50 (Suppl 1):**S41–S67.

Berggren S, Gall C, Wollnitz N, Ekelund M, Karlbom U, Hoogstraate J, Schrenk D, and Lennernäs H (2007) Gene and protein expression of P-glycoprotein, MRP1, MRP2, and CYP3A4 in the small and large human intestine. *Mol Pharm* **4:**252–257.

Brown RP, Delp MD, Lindstedt SL, Rhomberg LR, and Beliles RP (1997) Physiological parameter values for physiologically based pharmacokinetic models. *Toxicol Ind Health* **13:**407–484.

Chalasani N, Gorski JC, Patel NH, Hall SD, and Galinsky RE (2002) Reply to Rostami-Hodjegan and Tucker 2002; Hepatology 35(5): 1549–1550. *Hepatology* **35:**1550–1551.

Clifford CP, Adams DA, Murray S, Taylor GW, Wilkins MR, Boobis AR, and Davies DS (1997) The cardiac effects of terfenadine after inhibition of its metabolism by grapefruit juice. *Eur J Clin Pharmacol* **52:**311–315.

Darwich AS, Neuhoff S, Jamei M, and Rostami-Hodjegan A (2011) Interplay of metabolism and transport in determining oral drug absorption and gut wall metabolism: a simulation assessment using the "Advanced Dissolution, Absorption, Metabolism (ADAM)" model. *Curr Drug Metab* **11:**716–729.

Floren LC, Bekersky I, Benet LZ, Mekki Q, Dressler D, Lee JW, Roberts JP, and Hebert MF (1997) Tacrolimus oral bioavailability doubles with coadministration of ketoconazole. *Clin Pharmacol Ther* **62:**41–49.

Galetin A, Gertz M, and Houston JB (2008) Potential role of intestinal first-pass metabolism in the prediction of drug-drug interactions. *Expert Opin Drug Metab Toxicol* **4:**909–922.

Gammans RE, Mayol RF, and LaBudde JA (1986) Metabolism and disposition of buspirone. *Am J Med* **80 (3B):**41–51.

Gertz M, Davis JD, Harrison A, Houston JB, and Galetin A (2008a) Grapefruit juice-drug interaction studies as a method to assess the extent of intestinal availability: utility and limitations. *Curr Drug Metab* **9:**785–795.

Gertz M, Harrison A, Houston JB, and Galetin A (2010) Prediction of human intestinal first-pass metabolism of 25 CYP3A substrates from in vitro clearance and permeability data. *Drug Metab Dispos* **38:**1147–1158.

Gertz M, Kilford PJ, Houston JB, and Galetin A (2008b) Drug lipophilicity and microsomal protein concentration as determinants in the prediction of the fraction unbound in microsomal incubations. *Drug Metab Dispos* **36:**535–542.

Granger DN, Richardson PD, Kvietys PR, and Mortillaro NA (1980) Intestinal blood flow. *Gastroenterology* **78:**837–863.

Hebert MF, Fisher RM, Marsh CL, Dressler D, and Bekersky I (1999) Effects of rifampin on tacrolimus pharmacokinetics in healthy volunteers. *J Clin Pharmacol* **39:**91–96.

Hintz RJ and Johnson KC (1989) The effect of particle size distribution on dissolution rate and oral absorption. *Int J Pharm* **51:**9–17.

International Commission on Radiological Protection (2002) *ICRP Publication 89: Basic Anatomical and Physiological Data for Use in Radiological Protection: Reference Values* (Valetin J ed), Pergamon, Oxford, UK.

Jacobsen W, Kuhn B, Soldner A, Kirchner G, Sewing KF, Kollman PA, Benet LZ, and Christians U (2000) Lactonization is the critical first step in the disposition of the 3-hydroxy-3-methylglutaryl-CoA reductase inhibitor atorvastatin. *Drug Metab Dispos* **28:**1369–1378.

Jajoo HK, Mayol RF, LaBudde JA, and Blair IA (1989) Metabolism of the antianxiety drug buspirone in human subjects. *Drug Metab Dispos* **17:**634–640.

Jamei M, Turner D, Yang J, Neuhoff S, Polak S, Rostami-Hodjegan A, and Tucker G (2009) Population-based mechanistic prediction of oral drug absorption. *AAPS J* **11:**225–237.

Kameyama Y, Yamashita K, Kobayashi K, Hosokawa M, and Chiba K (2005) Functional characterization of SLCO1B1 (OATP-C) variants, SLCO1B1*5, SLCO1B1*15 and SLCO1B1*15+C1007G, by using transient expression systems of HeLa and HEK293 cells. *Pharmacogenet Genomics* **15:**513–522.

Kolars JC, Lown KS, Schmiedlin-Ren P, Ghosh M, Fang C, Wrighton SA, Merion RM, and Watkins PB (1994) CYP3A gene expression in human gut epithelium. *Pharmacogenetics* **4:**247–259.

Lampen A, Christians U, Bader A, Hackbarth I, and Sewing KF (1996) Drug interactions and interindividual variability of ciclosporin metabolism in the small intestine. *Pharmacology* **52:**159–168.

Lampen A, Christians U, Guengerich FP, Watkins PB, Kolars JC, Bader A, Gonschior AK, Dralle H, Hackbarth I, and Sewing KF (1995) Metabolism of the immunosuppressant tacrolimus in the small intestine: cytochrome P450, drug interactions, and interindividual variability. *Drug Metab Dispos* **23:**1315–1324.

Lennernäs H (2003) Clinical pharmacokinetics of atorvastatin. *Clin Pharmacokinet* **42:**1141–1160.

Mouly S and Paine MF (2003) P-glycoprotein increases from proximal to distal regions of human small intestine. *Pharm Res* **20:**1595–1599.

Nakamura T, Sakaeda T, Ohmoto N, Tamura T, Aoyama N, Shirakawa T, Kamigaki T, Nakamura T, Kim KI, Kim SR, et al. (2002) Real-time quantitative polymerase chain reaction for MDR1, MRP1, MRP2, and CYP3A-mRNA levels in Caco-2 cell lines, human duodenal enterocytes, normal colorectal tissues, and colorectal adenocarcinomas *Drug Metab Dispos* **30:**4–6.

Nestorov I (2003) Whole body pharmacokinetic models. *Clin Pharmacokinet* **42:**883–908.

Obach RS and Reed-Hagen AE (2002) Measurement of Michaelis constants for cytochrome P450-mediated biotransformation reactions using a substrate depletion approach. *Drug Metab Dispos* **30:**831–837.

Paine MF, Hart HL, Ludington SS, Haining RL, Rettie AE, and Zeldin DC (2006) The human intestinal cytochrome P450 "pie." *Drug Metab Dispos* **34:**880–886.

Paine MF, Khalighi M, Fisher JM, Shen DD, Kunze KL, Marsh CL, Perkins JD, and Thummel KE (1997) Characterization of interintestinal and intraintestinal variations in human CYP3A-dependent metabolism. *J Pharmacol Exp Ther* **283:**1552–1562.

Pasanen MK, Fredrikson H, Neuvonen PJ, and Niemi M (2007) Different effects of SLCO1B1 polymorphism on the pharmacokinetics of atorvastatin and rosuvastatin. *Clin Pharmacol Ther* **82:**726–733.

Poulin P and Theil FP (2002) Prediction of pharmacokinetics prior to in vivo studies. 1. Mechanism-based prediction of volume of distribution. *J Pharm Sci* **91:**129–156.

Prueksaritanont T, Subramanian R, Fang X, Ma B, Qiu Y, Lin JH, Pearson PG, and Baillie TA (2002) Glucuronidation of statins in animals and humans: a novel mechanism of statin lactonization. *Drug Metab Dispos* **30:**505–512.

Rodgers T, Leahy D, and Rowland M (2005) Physiologically based pharmacokinetic modeling 1: predicting the tissue distribution of moderate-to-strong bases. *J Pharm Sci* **94:**1259–1276.

Rodgers T and Rowland M (2006) Physiologically based pharmacokinetic modelling 2: predicting the tissue distribution of acids, very weak bases, neutrals and zwitterions. *J Pharm Sci* **95:**1238–1257.

Rostami-Hodjegan A and Tucker GT (2002) The effects of portal shunts on intestinal cytochrome P450 3A activity. *Hepatology* **35:**1549–1550; author reply 1550–1551.

Rostami-Hodjegan A and Tucker GT (2007) Simulation and prediction of in vivo drug metabolism in human populations from in vitro data. *Nat Rev Drug Discov* **6:**140–148.

Shen DD, Kunze KL, and Thummel KE (1997) Enzyme-catalyzed processes of first-pass hepatic and intestinal drug extraction. *Adv Drug Deliv Rev* **27:**99–127.

Strassburg CP, Barut A, Obermayer-Straub P, Li Q, Nguyen N, Tukey RH, and Manns MP (2001) Identification of cyclosporine A and tacrolimus glucuronidation in human liver and the gastrointestinal tract by a differentially expressed UDP-glucuronosyltransferase: UGT2B7. *J Hepatol* **34:**865–872.

Tam D, Tirona RG, and Pang KS (2003) Segmental intestinal transporters and metabolic enzymes on intestinal drug absorption. *Drug Metab Dispos* **31:**373–383.

Thörn M, Finnström N, Lundgren S, Rane A, and Lööf L (2005) Cytochromes P450 and MDR1 mRNA expression along the human gastrointestinal tract. *Br J Clin Pharmacol* **60:**54–60.

Thummel KE, O'Shea D, Paine MF, Shen DD, Kunze KL, Perkins JD, and Wilkinson GR (1996) Oral first-pass elimination of midazolam involves both gastrointestinal and hepatic CYP3A-mediated metabolism. *Clin Pharmacol Ther* **59:**491–502.

von Richter O, Burk O, Fromm MF, Thon KP, Eichelbaum M, and Kivistö KT (2004) Cytochrome P450 3A4 and P-glycoprotein expression in human small intestinal enterocytes and hepatocytes: a comparative analysis in paired tissue specimens. *Clin Pharmacol Ther* **75:**172–183.

Yang J, Jamei M, Yeo KR, Tucker GT, and Rostami-Hodjegan A (2007) Prediction of intestinal first-pass drug metabolism. *Curr Drug Metab* **8:**676–684.

Yeh KC, Stone JA, Carides AD, Rolan P, Woolf E, and Ju WD (1999) Simultaneous investigation of indinavir nonlinear pharmacokinetics and bioavailability in healthy volunteers using stable isotope labeling technique: study design and model-independent data analysis. *J Pharm Sci* **88:**568–573.

Yu LX and Amidon GL (1999) A compartmental absorption and transit model for estimating oral drug absorption. *Int J Pharm* **186:**119–125.

Yu LX, Lipka E, Crison JR, and Amidon GL (1996) Transport approaches to the biopharmaceutical design of oral drug delivery systems: prediction of intestinal absorption. *Adv Drug Deliv Rev* **19:**359–376.

Zhang QY, Dunbar D, Ostrowska A, Zeisloft S, Yang J, and Kaminsky LS (1999) Characterization of human small intestinal cytochromes P-450. *Drug Metab Dispos* **27:**804-809.

Zhu M, Zhao W, Jimenez H, Zhang D, Yeola S, Dai R, Vachharajani N, and Mitroka J (2005) Cytochrome P450 3A-mediated metabolism of buspirone in human liver microsomes. *Drug Metab Dispos* **33:**500–507.

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